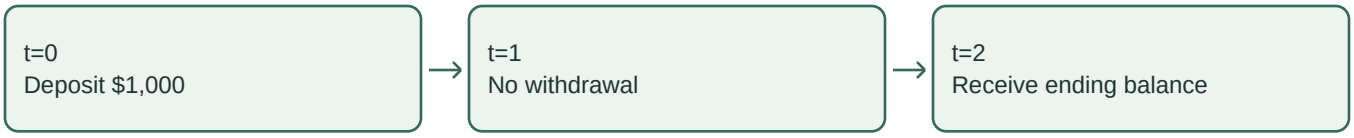
Cash-flow timelines & delayed payments



A. A single deposit earns 10% per year with annual compounding; no tax or fees.

A. Deposit / discounting

Starting deposit at t=0	$1,000.00
Annual effective rate	10%
Separate certain receipt at t=2	$1,210.00

The deposit is made once. Distinguish the ending balance from interest earned.

B. Two-payment contract

Payment received at t=1	$121.00
Payment received at t=2	$121.00
Annual effective rate	10%

No payment today. These are equal end-of-year payments. Value each cash flow at t=0 before combining them.

C. Endowment promise

Level annual payment	$50.00
First payment	One year from today
Last payment	None; payments continue forever
Discount rate	5% per year

Separate growth variation: first payment C1, growth rate g, discount rate r. The infinite growing-stream formula requires r > g. The level-payment case has g = 0.

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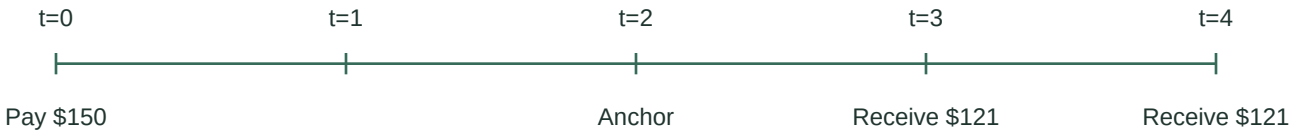

D. Delayed contract. Annual effective discount rate: 10%. No other cash flows.

D. Delayed contract inputs

Initial purchase price at t=0	$150.00
First receipt at t=3	$121.00
Second and final receipt at t=4	$121.00
Effective rate per year	10%

The ordinary-annuity formula values the receipts one period before the first payment, at t=2. Move that value back two more periods to today. Keep full precision until the final cent.

E. Timing and payment reference

Ordinary annuity	Equal payments at dates 1 through N
Annuity due	Equal payments at dates 0 through N-1
Payment on an amortizing loan	Principal / ordinary-annuity PV factor
Future savings goal	Target future value / ordinary-annuity FV factor
IRR	Rate making the signed cash-flow NPV zero

Cash-flow signs depend on whose perspective is used. A loan is positive at origination for the borrower and negative during repayment.

F. Tool and growth checks

Spreadsheet NPV(rate, future flows)	Discounts its first listed flow by one period
Cash flow today	Add separately; it is not discounted
Rate per period	Must match the spacing of payments
Finite growing stream	Can be discounted term by term even when g >= r

A calculator is optional. Multiplication, division, and a two-payment timeline are enough for the numerical koans in this chapter.